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Department of Mechanical Engineering

Discipline: Mining Machinery Engineering

Project Report

Design and Performance Analysis of a Semi-Active Quarter-Car Suspension System Using Magnetorheological (MR) Dampers

Simulation-based comparison of passive, fixed-current MR, Skyhook and PID suspension configurations

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Abstract

This report presents the design and simulation-based performance analysis of a semi-active quarter-car suspension that uses a magnetorheological (MR) damper. A two-degree-of-freedom model with a current-dependent MR damper force law was simulated in MATLAB (fourth-order Runge–Kutta, 1 ms step) for four configurations — passive damping, fixed-current MR damping at 0.5 A, Skyhook control and PID control — under eight road conditions: bumps of different height, bumps traversed at three speeds, and an ISO 8608-inspired Class C random road. Performance was compared using body displacement, peak and RMS body acceleration, suspension travel, tyre deflection, controller current and reported mechanical dissipated energy. Skyhook gave the lowest RMS body acceleration in seven of the eight cases (5.4–40.1% below passive) and PID in the remaining 10 km/h case (45.6% below passive), while PID gave the lowest body displacement in all eight cases. Suspension travel and tyre deflection favoured different configurations depending on the road input, and fixed 0.5 A MR damping did not consistently outperform passive damping. The results therefore show engineering trade-offs rather than a single superior controller. The study is simulation-only and relies on a simplified, uncalibrated damper model.

Keywords: quarter-car model; magnetorheological damper; semi-active suspension; Skyhook control; PID control; ride comfort; MATLAB simulation.

1. Introduction, Objectives and Project Stages

A vehicle suspension isolates the body from road-induced vibration while keeping the tyre in contact with the road. These goals conflict: soft damping lowers body acceleration (ride comfort) but permits large body and wheel motion, whereas firm damping controls that motion at the cost of transmitting more force to the body [5]. A passive spring–damper has fixed characteristics and is therefore a single compromise across road types, speeds and loads. Semi-active suspensions keep a passive spring but vary the damping in real time using very little control power, and can recover part of the benefit of fully active systems at much lower cost and complexity [1, 6]. MR dampers are well suited to this role: the damping force is changed electrically through coil current, with a response time of milliseconds and no mechanical control valves [3]. The project abstract identifies passenger, off-road and rail vehicles as target applications.

1.1 Objectives

- Develop a two-degree-of-freedom (2-DOF) quarter-car model with an MR damper as the semi-active element.
- Implement Skyhook and PID control strategies that vary the damper current within semi-active constraints.
- Simulate passive, fixed-current MR, Skyhook and PID configurations under bump, speed and random-road excitation.
- Quantify ride comfort (RMS body acceleration), suspension travel, tyre deflection and controller effort relative to a passive baseline.

1.2 Six-stage project overview

Table 1. The six completed project stages.

Stage	Title	Scope
1	**Quarter-Car Model Development**	Establishment of the two-degree-of-freedom quarter-car suspension model.
2	**MR Damper Modelling**	Development of the mathematical representation of the current-dependent MR damper.
3	**Control System Development**	Development of the Skyhook and PID control strategies.
4	**MATLAB Model Development**	Integration of the suspension model, MR damper and control system in MATLAB.
5	**Road Profile Simulation**	Simulation under bump and random-road excitation conditions using MATLAB (Stage5_RoadProfiles.m).
6	**Performance Analysis**	Comparison of passive and semi-active configurations using calculated performance metrics (final.m).

Stages 1–4 provide the model, damper and controllers used in this report; Stages 5 and 6 are discussed in detail below. All work reported here is MATLAB-based; a Simulink stage in the original plan was not undertaken.

2. Theoretical Background and System Modelling

2.1 MR fluids and Skyhook/PID control

MR fluid is a suspension of micron-sized ferromagnetic particles in a carrier oil. With no magnetic field it flows like a low-viscosity fluid; under a field the particles align into chains, producing a field-dependent yield stress usually described by a Bingham-type relation [3]. In an MR damper the coil current sets the field in the flow gap, so the force has a viscous part and a controllable yield (friction-like) part that both grow with current. Detailed hysteretic models such as the Spencer model [2] exist; this project uses a simplified algebraic form (Section 2.3). **Skyhook control** [1] attempts to apply a damping force proportional to the absolute body velocity, as if the body were damped against a fixed “sky” reference; since a real damper acts only between body and wheel, the demand can be met only when it opposes the relative velocity. **PID control** computes a force from a tracking error using proportional, integral and derivative terms; here the error is the body displacement from equilibrium.

2.2 Quarter-car system

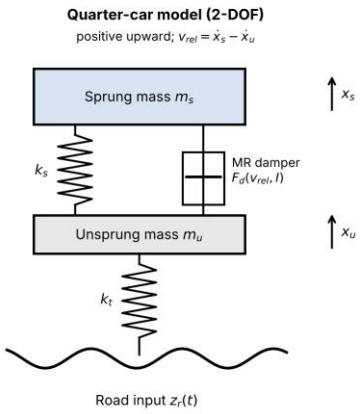


Fig. 1. Labelled quarter-car schematic.

The model represents one corner of the vehicle by a sprung mass m_s (body), an unsprung mass m_u (wheel assembly), a suspension spring k_s in parallel with the MR damper (force F_d) between them, and a linear tyre spring k_t connecting m_u to the road displacement $z_r(t)$; tyre damping is neglected. With x_s, x_u and z_r positive upward and relative velocity $v_{rel} = \dot{x}_s - \dot{x}_u$, the equations of motion are

$$m_s \ddot{x}_s = -k_s(x_s - x_u) - F_d \quad (1)$$

$$m_u \ddot{x}_u = k_s(x_s - x_u) + F_d - k_t(x_u - z_r) \quad (2)$$

Suspension travel is $x_s - x_u$ and tyre deflection is $x_u - z_r$. For the parameters in Table 2 the undamped natural frequencies are about 1.17 Hz (body mode) and 11.2 Hz (wheel-hop mode), computed from the 2-DOF stiffness and mass matrices.

2.3 MR damper model

The damper force used in the project is

$$F_d(v_{rel}, I) = [c_0 + c_s I]\, v_{rel} + F_{y,s}\, I \tanh(v_{rel}/v_\varepsilon) \quad (3)$$

where c_0 is the zero-current viscous coefficient, c_s the current sensitivity of the viscous coefficient, $F_{y,s}$ the yield-force sensitivity to current, I the coil current and v_ε a smoothing velocity that replaces $\mathrm{sgn}(v_{rel})$ to avoid a discontinuity at zero velocity. Both the viscous coefficient and the yield-like force therefore increase linearly with current. The actual current follows the commanded current through a first-order lag, with the command limited to 0–2 A and to a slew rate of 100 A/s:

$$dI/dt = (I_{cmd} - I)/\tau_I \quad (4)$$

This is a simplified algebraic (Bingham-type) damper with current lag; it is **not** the full Spencer phenomenological model [2] and its parameters have not been calibrated against experimental data. The passive reference uses a linear viscous damper, $F_d = c_p\, v_{rel}$.

2.4 Control strategies and configurations

Passive: fixed viscous damping c_p = 1000 N·s/m. **Fixed MR:** constant command of 0.5 A, giving a viscous coefficient of 1850 N·s/m and a yield-like term of 250 N·tanh(v_{rel}/v_ε); there is no feedback. **Skyhook:** the ideal demand is

$$F_{des} = -C_{sky}\, \dot{x}_s \quad (5)$$

applied only when $\dot{x}_s v_{rel}$ exceeds a hysteresis band of 0.002 m²/s² and $|v_{rel}|$ exceeds a 0.002 m/s deadband; otherwise the command is 0 A. **PID:** the error is $e = -x_s$ and

$$F_{des} = K_p\, e - K_d\, \dot{x}_s + K_i \int e\, dt \quad (6)$$

with the integral term bounded and frozen by conditional anti-windup. In both controllers the demanded force is converted to the current (searched over 0–2 A in 0.02 A steps) whose achievable damper force is closest to the demand; if the demand would require the damper to inject energy ($F_{des} v_{rel} > 0$) the current is set to 0 A. **PID force requests are therefore constrained by the semi-active damper** and are not applied as requested. Note that the damper at 0 A (c_0 = 800 N·s/m) is softer than the passive coefficient, so the controlled cases are softer than passive whenever near-zero current is commanded.

3. MATLAB Simulation Methodology and Parameters

3.1 Workflow

Stage5_RoadProfiles.m defines the eight road cases and four configurations, integrates the five-state system [x_s, $\dot{x}_s$, x_u, $\dot{x}_u$, I] with fixed-step fourth-order Runge–Kutta (Δt = 1 ms, zero initial conditions, controller command evaluated once per step and held through the RK4 sub-stages), computes the performance metrics for every case and configuration, and saves *Stage5_Results.mat* (time histories, metrics, case definitions, parameters). *final.m* loads this file, recalculates the metrics from the stored time histories, computes the percentage improvement over passive, identifies the best configuration per road test, evaluates controller current statistics and mechanical dissipated energy, checks that the damper power is never negative (minimum 0 W in all 32 simulations), generates the seven graphs and command-window outputs, exports four CSV tables and saves *Stage6_Results.mat*. The values in the MAT and CSV files were cross-checked and agree.

3.2 Simulation parameters

Table 2. Model, damper and controller parameters (values from Stage5_RoadProfiles.m).

Vehicle, tyre and simulation		MR damper		Controllers	
m_s	300 kg	c_0	800 N·s/m	C_{sky}	2500 N·s/m
m_u	40 kg	c_s	2100 N·s/(m·A)	Skyhook hysteresis	0.002 m²/s²
k_s	18 000 N/m	$F_{y,s}$	500 N/A	Velocity deadband	0.002 m/s
k_t	180 000 N/m	v_ε	0.005 m/s	K_p	18 000 N/m
c_p (passive)	1000 N·s/m	Current range	0–2 A (0.02 A grid)	K_i	1000 N/(m·s)
Time step Δt (RK4)	0.001 s	τ_I	0.03 s	K_d	3500 N·s/m
Bump length L	2 m	Max. slew rate	100 A/s	$\int e\,dt$ limit	±0.15 m·s
Bump start t_0	1 s	Fixed-MR current	0.5 A	Random-road seed	rng(1)

3.3 Road inputs

Table 3. The eight tested road conditions (from Stage5_RoadProfiles.m).

No.	Road test	Excitation	Height (mm)	Duration T_b (s)	Speed (km/h)	Record (s)
1	Baseline Bump	Half-sine bump	50	0.20	—	5
2	30 mm Bump	Half-sine bump	30	0.36	20	5
3	50 mm Bump †	Half-sine bump	50	0.36	20	5
4	80 mm Bump	Half-sine bump	80	0.36	20	5
5	Speed 10 km/h	Half-sine bump	50	0.72	10	5
6	Speed 20 km/h †	Half-sine bump	50	0.36	20	5
7	Speed 30 km/h	Half-sine bump	50	0.24	30	5
8	Random Road C	Random, ISO 8608-inspired Class C	—	—	20	10

† Cases 3 and 6 are the identical excitation (50 mm, 0.36 s) and give identical results; they are not independent tests.

Bumps are half-sine pulses starting at $t_0 = 1$ s:

$$z_r(t) = h \sin[\pi(t - t_0)/T_b], \quad 0 \le t - t_0 \le T_b \tag{7}$$

For the speed and height cases $T_b = L/v$ with $L = 2$ m; the baseline case uses a separately specified 0.20 s pulse. The random road is **ISO 8608-inspired Class C** (not formally ISO 8608-validated): a spatial profile built from 400 cosine components between 0.05 and 10 cycles/m with random phases, following

$$G_d(n) = G_0(n/n_0)^{-2}, \quad G_0 = 256\times10^{-6}\ \text{m}^3, \quad n_0 = 0.1\ \text{cycles/m} \tag{8}$$

and converted to a time signal at 20 km/h over a **10 s** record (zero initial displacement; RMS height 7.6 mm, computed from the saved profile).

3.4 Performance metrics

Metrics are evaluated for $t \ge 1$ s in bump cases and over the full 10 s in the random case: body displacement $\max|x_s|$; peak body acceleration $\max|\ddot{x}_s|$; RMS body acceleration; suspension travel $\max|x_s - x_u|$; and tyre deflection $\max|x_u - z_r|$ (a peak-value road-holding proxy, not dynamic tyre load). Body acceleration is $(-k_s(x_s - x_u) - F_d)/m_s$. Reported dissipated energy is $\int \max(F_d v_{rel}, 0)\, dt$ over the evaluation window (mechanical damper dissipation). Improvement relative to passive is

$$\text{Improvement (\%)} = [(\text{Passive value} - \text{Controlled value}) / \text{Passive value}] \times 100 \tag{9}$$

so positive values indicate a reduction of the metric and negative values a deterioration.

4. Results and Graphical Analysis

4.1 Absolute performance

Tables 4–7 and Figs. 2–4 reproduce the supplied Stage 6 outputs. Skyhook has the lowest RMS body acceleration in seven of the eight cases; PID is lowest at 10 km/h (0.462 vs 0.509 m/s² for Skyhook). Skyhook also has the lowest peak body acceleration in all eight cases. PID has the lowest body displacement in all eight cases (e.g. 38.4 mm vs 60.2 mm for passive at the 50 mm bump, and 62.7 vs 96.3 mm at 80 mm). Suspension travel and tyre deflection have no single best configuration (Table 7): fixed 0.5 A MR damping gives the least travel in the baseline, 10 km/h and random cases, PID in the 30, 50 and 80 mm bumps and at 20 km/h, and Skyhook (tied with PID) at 30 km/h; the passive damper gives the least tyre deflection at 10 km/h and on the random road, and Skyhook in the other six cases.

Table 4. Performance summary (Stage6_Performance_Summary.csv). Bold = lowest value of that metric within a road test; acceleration in m/s².

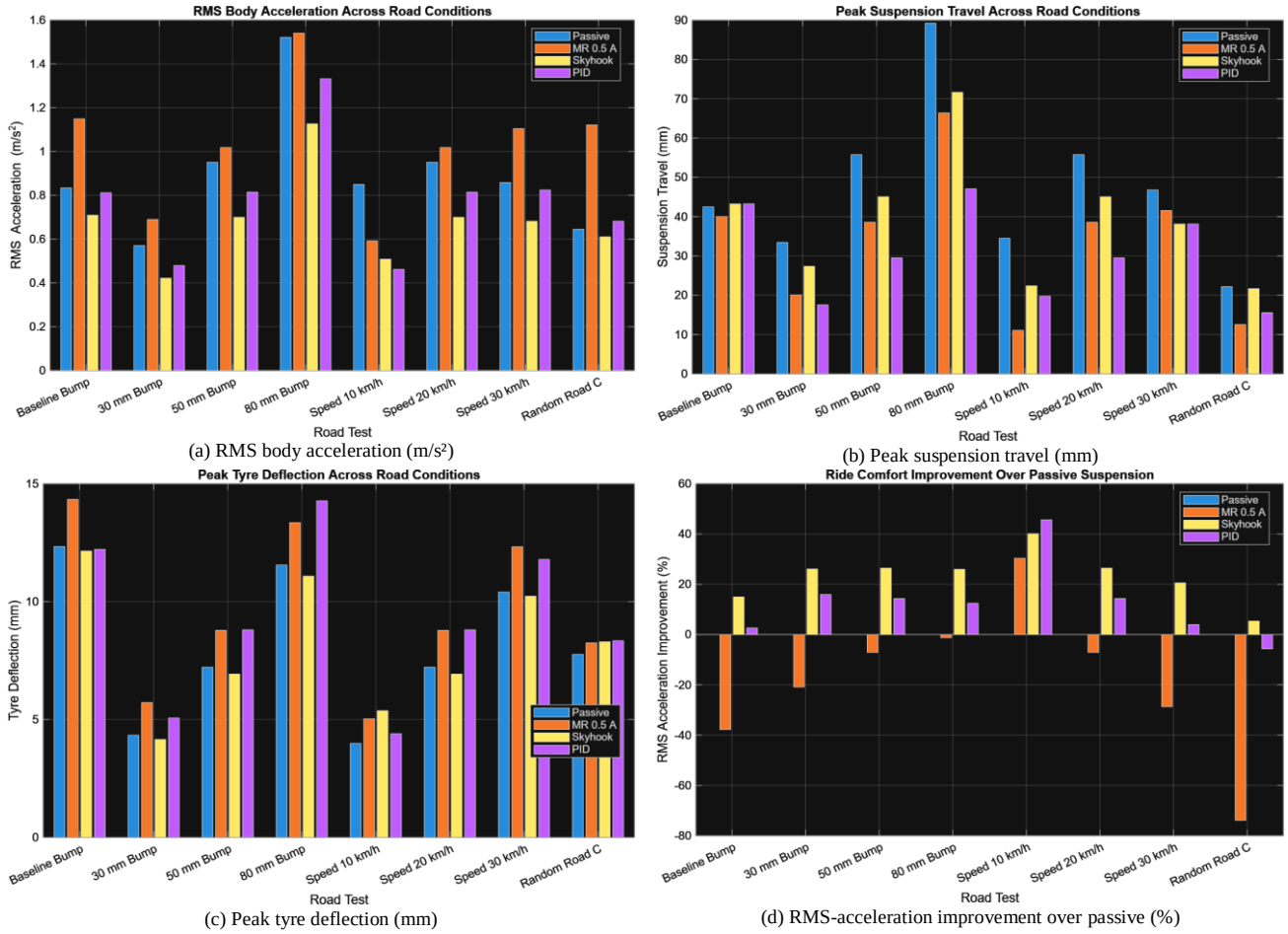
Road test	System	Disp. (mm)	Peak acc.	RMS acc.	Travel (mm)	Tyre (mm)	Road test	System	Disp. (mm)	Peak acc.	RMS acc.	Travel (mm)	Tyre (mm)
Baseline Bump	Passive	38.04	4.786	0.834	42.50	12.34	Speed 10 km/h	Passive	72.65	2.244	0.850	34.51	**3.99**
	MR 0.5 A	41.94	7.168	1.149	**40.02**	14.34		MR 0.5 A	55.38	2.438	0.592	**11.07**	5.03
	Skyhook	31.98	**4.327**	**0.709**	43.24	**12.15**		Skyhook	50.67	**2.170**	0.509	22.40	5.37
	PID	**26.16**	5.080	0.812	43.24	12.22		PID	**35.36**	2.257	**0.462**	19.76	4.40
30 mm Bump	Passive	36.11	2.546	0.571	33.46	4.33	Speed 20 km/h	Passive	60.18	4.243	0.951	55.77	7.22
	MR 0.5 A	33.71	2.742	0.690	20.07	5.72		MR 0.5 A	56.14	4.221	1.018	38.60	8.78
	Skyhook	29.76	**2.049**	**0.422**	27.39	**4.16**		Skyhook	49.06	**3.379**	**0.700**	45.07	**6.93**
	PID	**22.80**	2.380	0.480	**17.56**	5.07		PID	**38.37**	4.172	0.815	**29.50**	8.80
50 mm Bump	Passive	60.18	4.243	0.951	55.77	7.22	Speed 30 km/h	Passive	44.90	4.484	0.858	46.80	10.41
	MR 0.5 A	56.14	4.221	1.018	38.60	8.78		MR 0.5 A	46.64	6.252	1.104	41.57	12.33
	Skyhook	49.06	**3.379**	**0.700**	45.07	**6.93**		Skyhook	33.67	**3.684**	**0.682**	**38.13**	**10.23**
	PID	**38.37**	4.172	0.815	**29.50**	8.80		PID	**26.42**	4.877	0.824	**38.13**	11.79
80 mm Bump	Passive	96.29	6.788	1.521	89.23	11.55	Random Road C	Passive	24.31	2.180	0.645	22.18	**7.76**
	MR 0.5 A	90.15	6.702	1.540	66.42	13.35		MR 0.5 A	24.74	3.358	1.121	**12.54**	8.25
	Skyhook	78.28	**5.372**	**1.126**	71.66	**11.08**		Skyhook	23.74	**2.006**	**0.611**	21.69	8.30
	PID	**62.72**	6.859	1.332	**47.09**	14.28		PID	**22.56**	2.450	0.682	15.57	8.34

4.2 Improvement relative to passive suspension

Skyhook reduces RMS acceleration in every case (5.4% on the random road, 15.0% for the baseline bump, 20.6% at 30 km/h, 26.0–26.4% for the 30/50/80 mm and 20 km/h cases, and 40.1% at 10 km/h) and also reduces peak acceleration in every case (3.3–20.9%). PID reduces RMS acceleration in all bump and speed cases (2.7–45.6%) but increases it by 5.7% on the random road, and its peak acceleration ranges from 12.4% worse to 6.5% better than passive. Fixed 0.5 A MR damping increases RMS acceleration in seven of the eight cases (1.2–73.8%) and improves it only at 10 km/h (30.3%); it reduces suspension travel in all cases (5.8–67.9%) but increases tyre deflection in all cases (6.2–32.1%). Skyhook lowers tyre deflection in six cases (1.5–4.1%) but raises it at 10 km/h (34.8%) and on the random road (6.9%); PID improves it only for the baseline bump (1.0%) and worsens it by 7.4–23.6% elsewhere. PID's body-displacement reduction is 7.2–51.3%, against 2.3–30.3% for Skyhook.

Table 5. Improvement over passive suspension, % (Stage6_Improvements.csv). Positive = reduction in the metric; negative values in red. Passive rows (0 by definition) omitted.

Road test	System	Disp.	Peak acc.	RMS acc.	Travel	Tyre	Road test	System	Disp.	Peak acc.	RMS acc.	Travel	Tyre
Baseline Bump	MR 0.5 A	−10.3	−49.8	−37.8	5.8	−16.2	Speed 10 km/h	MR 0.5 A	23.8	−8.7	30.3	67.9	−26.1
	Skyhook	15.9	9.6	15.0	−1.8	1.5		Skyhook	30.3	3.3	40.1	35.1	−34.8
	PID	31.2	−6.2	2.7	−1.8	1.0		PID	51.3	−0.6	45.6	42.7	−10.3
30 mm Bump	MR 0.5 A	6.6	−7.7	−20.9	40.0	−32.1	Speed 20 km/h	MR 0.5 A	6.7	0.5	−7.1	30.8	−21.6
	Skyhook	17.6	19.5	26.1	18.1	4.1		Skyhook	18.5	20.4	26.4	19.2	4.1
	PID	36.9	6.5	15.9	47.5	−16.9		PID	36.2	1.7	14.3	47.1	−21.9
50 mm Bump	MR 0.5 A	6.7	0.5	−7.1	30.8	−21.6	Speed 30 km/h	MR 0.5 A	−3.9	−39.4	−28.7	11.2	−18.5
	Skyhook	18.5	20.4	26.4	19.2	4.1		Skyhook	25.0	17.9	20.6	18.5	1.6
	PID	36.2	1.7	14.3	47.1	−21.9		PID	41.2	−8.8	4.0	18.5	−13.3
80 mm Bump	MR 0.5 A	6.4	1.3	−1.2	25.6	−15.6	Random Road C	MR 0.5 A	−1.8	−54.0	−73.8	43.5	−6.2
	Skyhook	18.7	20.9	26.0	19.7	4.1		Skyhook	2.3	8.0	5.4	2.2	−6.9
	PID	34.9	−1.0	12.5	47.2	−23.6		PID	7.2	−12.4	−5.7	29.8	−7.4



(a) RMS body acceleration (m/s²)

(b) Peak suspension travel (mm)

(c) Peak tyre deflection (mm)

(d) RMS-acceleration improvement over passive (%)

Fig. 2. Stage 6 comparison charts across the eight road conditions: (a) RMS body acceleration, (b) peak suspension travel, (c) peak tyre deflection, (d) ride-comfort improvement over passive suspension. Cases 3 and 6 are identical excitations.

4.3 Controller current and mechanical energy

Skyhook and PID idle near zero current for most of the record: average current is 0.026–0.071 A for Skyhook and 0.039–0.135 A for PID, against a constant 0.5 A for the fixed MR case (Fig. 4). Maximum currents are 0.40–1.12 A (Skyhook) and 0.71–1.94 A (PID), below the 2 A limit. PID draws more average current than Skyhook in every case, and both peak at 10 km/h, the longest (0.72 s) bump. Reported mechanical dissipated energy (Table 6) is lower than passive for Skyhook in all eight cases (0.4–28.8% lower) and for PID in five (e.g. 11.7 J vs 40.3 J at 10 km/h), but higher for PID on the baseline bump, at 30 km/h and on the random road. Fixed 0.5 A MR damping dissipates 26–62% more than passive in six cases and less at 10 km/h and on the random road. These figures are mechanical dissipation in the damper, **not** electrical energy consumption.

Table 6. Controller current and reported mechanical dissipated energy (Stage6_Controller_Usage.csv). Current statistics over the evaluation window (maximum over the full record). The CSV's $\int I\,dt$ and $\int I^2\,dt$ columns are omitted; minimum damper power is 0 W in every case.

Road test	System	Avg *I* (A)	RMS *I* (A)	Max *I* (A)	E_{diss} (J)	Road test	System	Avg *I* (A)	RMS *I* (A)	Max *I* (A)	E_{diss} (J)
Baseline Bump	Passive	0.0000	0.0000	0.0000	108.01	**Speed 10 km/h**	Passive	0.0000	0.0000	0.0000	40.35
	MR 0.5 A	0.5000	0.5000	0.5000	175.44		MR 0.5 A	0.5000	0.5000	0.5000	20.71
	Skyhook	0.0297	0.0799	0.4704	107.47		Skyhook	0.0711	0.2012	1.1230	28.75
	PID	0.0393	0.1195	0.7094	132.22		PID	0.1350	0.4083	1.9421	11.65
30 mm Bump	Passive	0.0000	0.0000	0.0000	28.17	**Speed 20 km/h**	Passive	0.0000	0.0000	0.0000	78.24
	MR 0.5 A	0.5000	0.5000	0.5000	38.80		MR 0.5 A	0.5000	0.5000	0.5000	101.99
	Skyhook	0.0323	0.0847	0.4273	25.57		Skyhook	0.0461	0.1189	0.6217	70.80
	PID	0.0503	0.1561	0.9864	26.26		PID	0.0712	0.2173	1.3651	73.77
50 mm Bump	Passive	0.0000	0.0000	0.0000	78.24	**Speed 30 km/h**	Passive	0.0000	0.0000	0.0000	92.93
	MR 0.5 A	0.5000	0.5000	0.5000	101.99		MR 0.5 A	0.5000	0.5000	0.5000	149.18
	Skyhook	0.0461	0.1189	0.6217	70.80		Skyhook	0.0376	0.0991	0.6127	91.29
	PID	0.0712	0.2173	1.3651	73.77		PID	0.0600	0.1940	1.4149	112.29
80 mm Bump	Passive	0.0000	0.0000	0.0000	200.30	**Random Road C**	Passive	0.0000	0.0000	0.0000	244.48
	MR 0.5 A	0.5000	0.5000	0.5000	252.54		MR 0.5 A	0.4984	0.4988	0.5000	232.17
	Skyhook	0.0624	0.1557	0.8260	182.34		Skyhook	0.0262	0.0555	0.3992	243.62
	PID	0.0947	0.2738	1.6388	193.19		PID	0.0825	0.1431	0.7771	248.35

4.4 Best configurations and random-road response

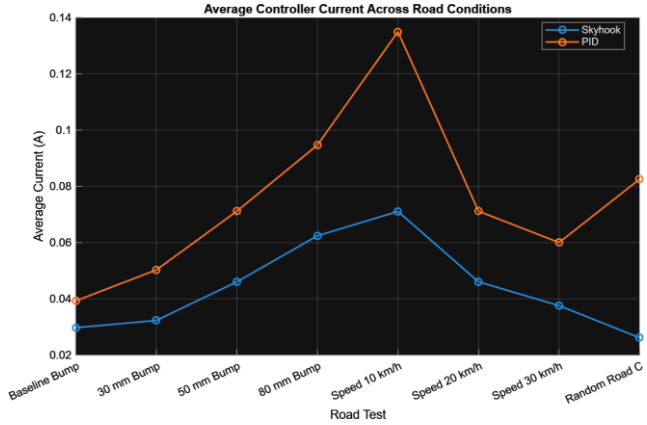
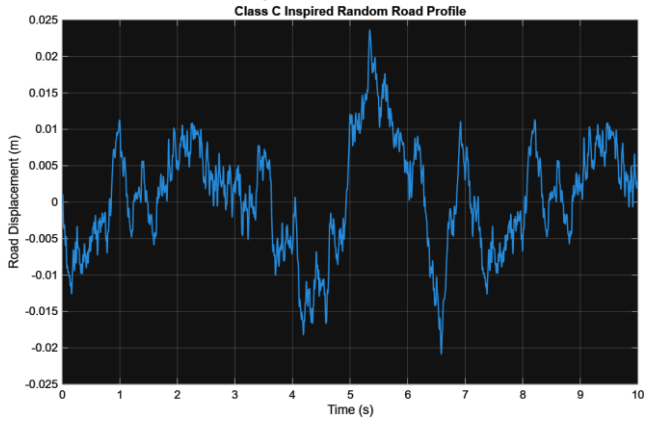


Fig. 4. Average controller current (Skyhook, PID) across road conditions.

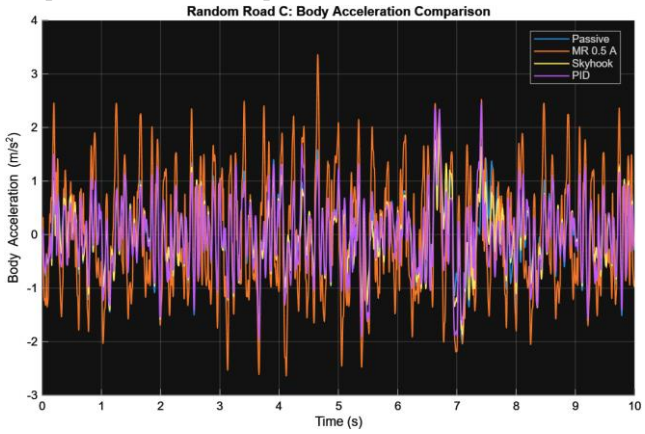
Table 7. Best configuration per road test (Stage6_Best_Configurations.csv). *Skyhook and PID tie (38.13 mm).

Road test	Best comfort	Lowest travel	Lowest tyre	Min. RMS (m/s²)
Baseline Bump	Skyhook	MR 0.5 A	Skyhook	0.709
30 mm Bump	Skyhook	PID	Skyhook	0.422
50 mm Bump	Skyhook	PID	Skyhook	0.700
80 mm Bump	Skyhook	PID	Skyhook	1.126
Speed 10 km/h	PID	MR 0.5 A	Passive	0.462
Speed 20 km/h	Skyhook	PID	Skyhook	0.700
Speed 30 km/h	Skyhook	Skyhook*	Skyhook	0.682
Random Road C	Skyhook	MR 0.5 A	Passive	0.611

On the random road (Fig. 3), fixed 0.5 A MR damping shows visibly larger acceleration peaks (RMS 1.121 vs 0.645 m/s² passive; peak 3.358 vs 2.180 m/s²). Skyhook gives the lowest RMS (0.611 m/s²) and peak (2.006 m/s²) acceleration, 5.4% and 8.0% below passive; PID gives the lowest body displacement (22.6 vs 24.3 mm) but 5.7% higher RMS acceleration. All three MR-based configurations show slightly higher tyre deflection than passive (8.25–8.34 vs 7.76 mm). The gains on the random road are small compared with the bump cases.



(a) Class C inspired random road profile



(b) Body acceleration comparison

Fig. 3. Random Road C (20 km/h, 10 s): (a) road displacement profile; (b) body acceleration of the four configurations.

5. Discussion and Limitations

5.1 Engineering trade-offs

No configuration is best on every metric. **Skyhook** reacts to absolute body velocity, which explains its lowest RMS and peak body acceleration; it also needs little current, but its reductions in body displacement and travel are more modest and it raises tyre deflection at 10 km/h and on the random road. **PID** regulates body displacement, which explains its lowest displacement in all cases and large travel reductions in most bump cases (42.7–47.5% in cases 2–6; 29.8% on the random road); because its demand is displacement-based and feasible only when it opposes the relative velocity, its peak accelerations are often no better than passive and it draws more current. **Fixed 0.5 A** MR damping is a stiffer damper (1850 N·s/m plus a 250 N yield-like term): it strongly limits travel but raises acceleration and tyre deflection in most cases, so a fixed current is not a substitute for control. The **passive** damper retains the lowest tyre deflection at 10 km/h and on the random road. The 10 km/h case (0.72 s bump) favours PID, whose RMS reduction (45.6%) exceeds Skyhook's (40.1%). Skyhook and PID travel values coincide for the baseline bump and 30 km/h because the peak occurs at about 1.08 s, when both controllers are still commanding 0 A (checked in the stored time histories). The softer zero-current damping (800 vs 1000 N·s/m) likely contributes to some of the differences from passive.

Expected outcomes versus results. The project abstract set benchmark-based expectations from published studies of about 35% lower RMS body acceleration, 22% lower suspension travel and 28% better tyre–road contact. These were expectations, not results of this work. In the present simulation, an RMS reduction of at least 35% occurs only at 10 km/h (Skyhook 40.1%, PID 45.6%; the best controller achieves up to 26.4% in the other cases); a travel reduction of at least 22% is reached by PID or fixed MR in six of eight cases (not the baseline bump, 5.8%, or 30 km/h, 18.5%); and tyre deflection improves by at most 4.1%, far short of the 28% expectation.

5.2 Limitations

- The MR damper is a simplified algebraic model (Eq. 3) with current lag; it has no experimental calibration and omits hysteresis, temperature and saturation effects.

- The random road is ISO 8608-inspired, a single realisation (rng(1)), and was not formally validated against ISO 8608.
- The 50 mm bump and 20 km/h cases are identical, so the eight cases contain only seven independent excitations.
- A quarter-car model neglects pitch, roll and load transfer; no full-vehicle model was used.
- There is no physical prototype or experimental validation; all results are simulation-only.
- No fully active suspension was simulated, so the claim of approaching active performance is not tested.
- Electrical energy consumption cannot be established from current alone (coil resistance and drive electronics are not modelled); the reported dissipated energy is mechanical damper dissipation, not electrical consumption.
- Controller gains were not optimised, and the metrics are peak or RMS values over finite windows.

6. Conclusion and Future Scope

A 2-DOF quarter-car model with a current-dependent MR damper was simulated under eight road conditions for passive, fixed 0.5 A, Skyhook and PID configurations. Skyhook gave the lowest RMS body acceleration in seven of eight cases and the lowest peak acceleration in all eight, with low current demand; PID gave the lowest RMS acceleration at 10 km/h and the lowest body displacement in every case; fixed 0.5 A damping reduced travel but did not consistently outperform passive damping. Suspension travel and tyre deflection had road-dependent best configurations, so the preferred strategy depends on whether comfort, body control or road holding is prioritised. These conclusions apply to the simulated model only.

Future work: experimental force–velocity–current characterisation of an MR damper and parameter identification; controller optimisation (gain tuning and combined or multi-objective Skyhook–PID strategies, including tyre deflection); advanced hysteretic damper models; full-vehicle simulation; and the physical proof-of-concept test rig outlined in the project plan.

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